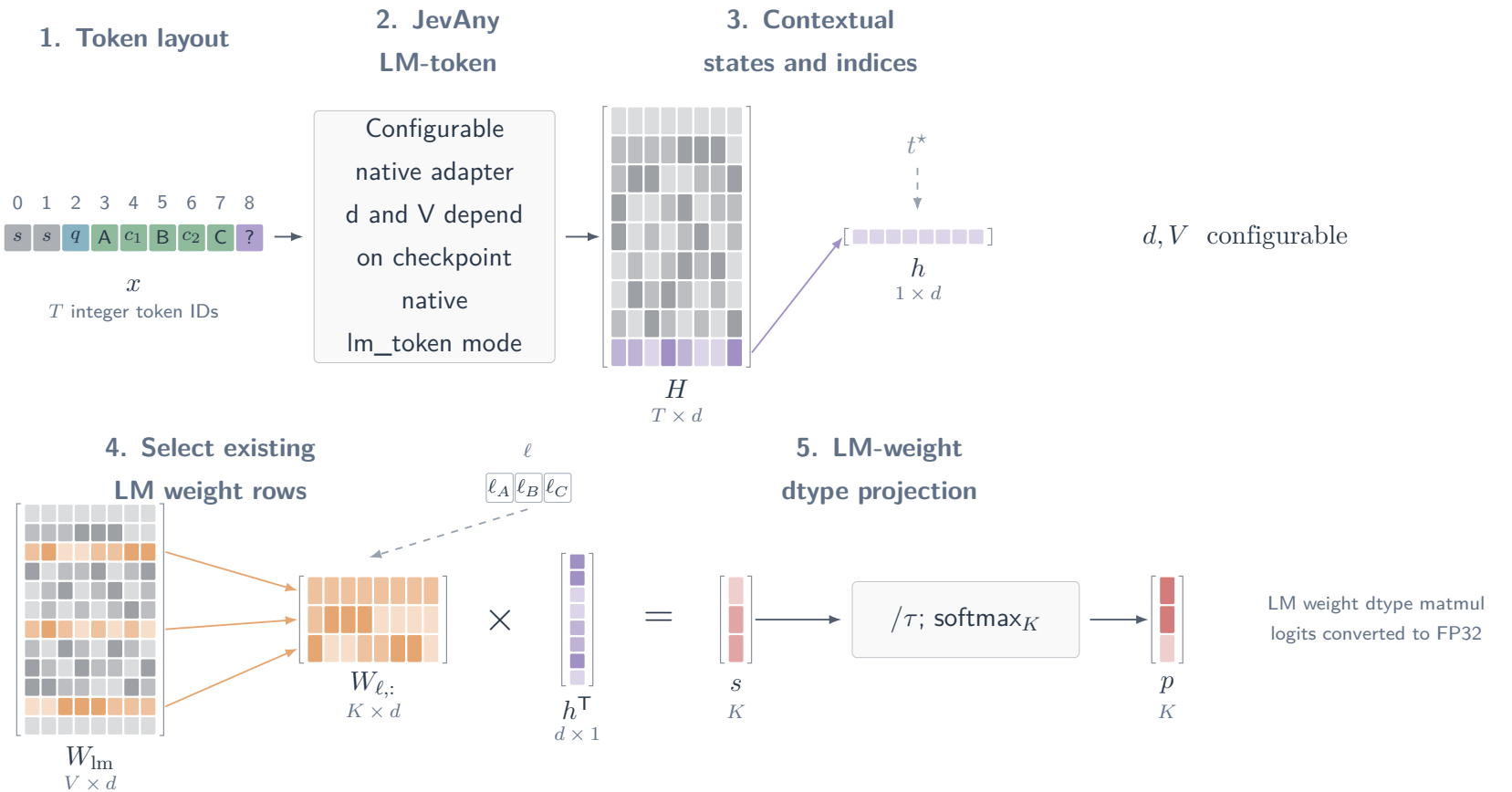
$$h = H_{t^\star,:}, \quad s_k = \langle W_{\ell_k,:}, h \rangle, \quad p = \mathrm{softmax}_K(s/\tau)$$



Axes	d, V configurable; Single-question illustration; batch omitted. Tile counts encode axes, not literal sizes.
Objects	Gray: input/context; violet: hidden states; orange: parameter matrices; red: logits/probabilities. Positions, code IDs and owner maps are integers, not scores. Tiles show roles, not measured magnitudes.
Mechanism	One distinct single-token verbalizer per choice. Training-free choice with alias-logsumexp is another path; it is not mixed into this graph. Native training uses full-vocabulary CE at the decision position with no EOS target. Inference selects existing frozen LM rows before projection.